

JUNSU KIM

Curriculum Vitae

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Research Interests

- Egocentric and embodied AI for tool use, physical interactions, and action-driven state changes.
 - 3D/4D scene understanding and action-centric world models trained on human and robot data.
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Education

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

B.S. in Electrical Engineering

Mar 2021 – Expected Aug 2027

GPA: 4.09/4.3 (Major GPA: 4.19/4.3)

Military Service, Republic of Korea Army: Jul 2022 – Jan 2024

Nanyang Technological University (NTU)

Singapore

Exchange Student

Jan 2026 – May 2026

School of Electrical and Electronic Engineering

Publications

[1] EgoTools: Towards Tool-Centric Reasoning in Real-World Egocentric Videos.

[\[Project Page\]](#)

Shulin Tian*, Junsu Kim*, Shuai Liu*, Hao Li*, Yujiao Shen, Sihan Li, Zhe Yang, Yeongon Kim, Feiyu Li, Jialin Wu, Yichi Zhang, Wenhui Wang, Runmao Yao, Yuhao Dong, Zhaoxi Chen, Fangzhou Hong, Antonino Furnari, Jingkang Yang, Hongyuan Zhu, Ziwei Liu.

EMNLP 2026 (Main Conference).

Contribution: Co-led dataset and benchmark development, designed annotation and QA workflows, constructed the 1,000-question benchmark, fine-tuned Qwen3-VL-8B, and conducted baseline and generalization analyses.

[2] C3G: Learning Compact 3D Representations with 2K Gaussians.

[\[Project Page\]](#) [\[Paper\]](#).

Honggyu An*, Jaewoo Jung*, ..., Hyuna Ko, Junsu Kim, ..., Seungryong Kim.

CVPR 2026.

Contribution: Implemented analysis tools and conducted experiments for feed-forward 3D reconstruction pipelines, focusing on novel-view synthesis quality and compact 3D representation behavior.

*Equal contribution.

Research Experience

MAUM AI, WoRV Division, Future Intelligence Lab

Aug 2026 – Present

Research Intern

- Evaluating real-world vision-language-action and world-action models and developing human-aware embodied evaluation tasks, with a focus on physical reasoning and action-conditioned behaviors.

S-Lab (MMLab), NTU

Jan 2026 – Present

Research Intern (Advisor: [Prof. Ziwei Liu](#), NTU)

- Co-led **EgoTools**, a real-world egocentric dataset and benchmark for tool-centric reasoning, built from **100+ hours** of long-horizon tool-use videos.
- Designed annotation and QA workflows for physical-state, procedural, affordance, and tool-use reasoning; coordinated data collection and quality control; and co-led construction of a **1K-QA** benchmark.
- Fine-tuned **Qwen3-VL-8B** on **220K** multimodal training examples, improving EgoTools-Bench accuracy from **50.0%** to **60.9%**; conducted cross-benchmark generalization, failure-case, and catastrophic-forgetting analyses.

Computer Vision Lab (CVLab), KAIST AI

Jun 2025 – Present

Research Intern (Advisor: [Prof. Seungryong Kim](#), KAIST AI)

- *Jun 2026 – Present*: Investigating latent action representations for video dynamics modeling and action-conditioned future prediction.
- *Jun 2025 – Dec 2025*: Conducted feed-forward 3D reconstruction and novel-view synthesis research using Gaussian Splatting, VGGT, and NoPoSplat, focusing on geometry quality, failure cases, and compact 3D representation behavior.

Video and Image Computing Lab (VICLab), KAIST

Dec 2024 – Jun 2025

Research Intern (Advisor: [Prof. Mun Churl Kim](#), KAIST EE)

- Implemented and evaluated object-detection pipelines spanning two-stage, YOLO-style, and Vision Transformer architectures for few-shot and anomaly detection tasks.

Honors & Scholarships

KAIST Presidential Fellowship (15th)

Feb 2025 – Present

National Science & Engineering Scholarship for Academic Excellence

Aug 2025 – Present

Woon Hae Scholarship (12th)

Feb 2025 – Dec 2025

KAIST Dean's List, College of Engineering

Fall 2021, Spring 2024

awarded to the top 3% of students

KAIST Shared Lives Award (Top-ranked Recipient)

Feb 2026

For exceptional and sustained blood donation service

Leadership & Service

President, EERun (KAIST EE Running Club)

Sep 2024 – Dec 2025

Organized weekly training sessions and semester events for KAIST EE students.

Google Student Ambassador

Sep 2025 – Dec 2025

Promoted Gemini Pro to KAIST students through short-form video content.

Additional Activities

2022 – Present

KAIST AI Studying Club, Young Engineers Honor Society (YEHS), SilverLining, Freshman Proctor.

Skills

Programming

C/C++, Python, PyTorch, Git, LaTeX

Research Tools

video preprocessing, dataset curation, annotation pipelines, benchmark construction, evaluation scripts

Languages

Korean (native), English (fluent, iBT TOEFL 110)